Yifan Zeng

Ph.D. Student · RL for LLM Agents

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Education

Oregon State University — Corvallis, OR
Ph.D. in Computer Science — 2023–Spring 2028 *(expected)*

- Advisor: Prof. Huazheng Wang
- Research Areas: RL for LLM Agents, Multi-Agent Systems, LLM Robustness
- Publications: 12 papers (6 first/co-first author), incl. NeurIPS, ICLR, EMNLP
- Service: Reviewer for ICLR, NeurIPS, AAAI, and WWW (2025–2027)

Zhuhai College of Jilin University — Zhuhai, China
B.E. in Data Science — 2019–2023

- GPA: 3.84/4.00, Rank: 1/249

Industry Experience

Research Scientist Intern | Adobe — Jun 2026 – Sep 2026

- Developed SkillSaw, a training-free search over how procedural knowledge is divided into agent skills: it regroups verifiable knowledge atoms and rewrites skill descriptions from routing and task-success evidence (paper under review)
- Significantly raised held-out success on 7 of 9 benchmark–model pairs (none regressed), e.g. AppWorld 79.9% → 95.7%, LiveMath 33.8% → 56.0%
- Gains track receipt: the AppWorld skill holding a key task rule is opened in 97.9% of test episodes (up from 12.8%) while each episode loads a small share of the library
- Applied to Adobe Experience Platform Agent Orchestrator and other agent harnesses

AI Agent Research Intern | AG2AI — Jun 2025 – Sep 2025

- Co-first author of SimpleDoc (EMNLP 2025 Main): a document-QA agent that retrieves pages with visual embeddings plus LLM re-ranking over page summaries, then iteratively refines its query using a working memory
- Outperformed prior baselines by 3.2% on average across 4 DocVQA benchmarks while reading about 3.5 pages per question; open-sourced as ag2ai/SimpleDoc

Applied Scientist | Quanbao Network Technology — Apr 2022 – Aug 2023

- Built self-supervised pretraining and object detection for bronchoscopy landmark detection (deployed on endoscopic devices) and railway ultrasonic defect inspection

Academic Experience

Oregon State University — Mar 2024 – Present
Graduate Research Assistant — Mar 2024 – Present

- **Multi-agent RL for LLM workflows** (first author): built GRPO training on verl with per-role LoRA adapters and vLLM rollouts; trained 36 workflow configurations and 6 single-agent baselines on Qwen3 0.6B–4B for math and code (about 235 node-days), showing when multi-agent training beats single-agent RL in the same workflow (evaluator–optimizer with per-role policies: 3.3–9.1 points on DAPO-Math)
- **LLM alignment**: co-first author of an analysis of failure modes in margin-based preference optimization (ICLR 2025); built RLHF pipelines (SFT, reward modeling, preference optimization) with TRL
- **Safety and robustness**: AutoDefense multi-agent jailbreak defense (featured on the AutoGen blog), prompt-injection attacks on LLM rankers (EMNLP 2025 Main), and LLM-RankFusion for ranking consistency (TMLR)

Graduate Teaching Assistant — Sep 2024 – Jun 2025, Sep 2025 – Present

- TA for graduate and undergraduate CS courses; mentored AI Capstone teams building LLM-agent RAG products for funding and patent search

Skills

LLM Post-Training & RL: SFT, RLHF, Preference Optimization (DPO), Online RL (GRPO, PPO) for Tool-Using, Long-Horizon, and Multi-Agent Workflows, Reward Design and Modeling, LoRA, Multi-Node Distributed Training (DP/TP), Inference-Time Scaling (Best-of-N, Tree Search)

LLM Agents: Multi-Agent Workflows, Agent Harness and Skill-Library Optimization, Agent Memory, Tool Use (MCP), Long-Horizon Tasks, Agent and Prompt Optimization

Evaluation & Safety: Agent Benchmarks, Trajectory Evaluation, Failure Attribution, LLM-as-Judge, Jailbreak Defense, Prompt Injection

Retrieval & Ranking: LLM Rerankers, Rank Aggregation, Multi-Modal Document RAG

Frameworks: PyTorch, verl, Hugging Face (Transformers, TRL, PEFT), vLLM, Ray, Docker

Selected Publications

First or Co-First Author

1. **Yifan Zeng**, Yiran Wu, Yaolun Zhang, Wentian Zhao, Kun Wan, Qingyun Wu, Huazheng Wang. **When Does Multi-Agent RL Improve LLM Workflows? Workflow, Scale, and Policy-Sharing Tradeoffs**, *arXiv preprint arXiv:2605.24202*, 2026.
2. Chelsi Jain*, Yiran Wu*, **Yifan Zeng***, Jiale Liu, Shengyu Dai, Zhenwen Shao, Qingyun Wu, Huazheng Wang. **SimpleDoc: Multi-Modal Document Understanding with Dual-Cue Page Retrieval and Iterative Refinement**, EMNLP 2025 (Main Conference).
3. Yaoyao Qian*, **Yifan Zeng***, Yuchao Jiang, Chelsi Jain, Huazheng Wang. **The Ranking Blind Spot: Decision Hijacking in LLM-based Text Ranking**, EMNLP 2025 (Main Conference).
4. Hui Yuan*, **Yifan Zeng***, Yue Wu*, Huazheng Wang, Mengdi Wang, Liu Leqi*. **A Common Pitfall of Margin-based Language Model Alignment: Gradient Entanglement**, ICLR 2025.
5. **Yifan Zeng***, Ojas Tendolkar*, Raymond Baartmans*, Qingyun Wu, Lizhong Chen, Huazheng Wang. **LLM-RankFusion: Mitigating Intrinsic Inconsistency in LLM-based Ranking**, Transactions on Machine Learning Research (TMLR), 2024.
6. **Yifan Zeng***, Yiran Wu*, Xiao Zhang, Huazheng Wang, Qingyun Wu. **AutoDefense: Multi-Agent LLM Defense against Jailbreak Attacks**, NeurIPS 2024 Workshop SafeGenAI.

Co-Author

7. Jiale Liu, Huajun Xi, Shaokun Zhang, **Yifan Zeng**, Tianwei Yue, Chi Wang, Jian Kang, Qingyun Wu, Huazheng Wang. **Who&When Pro: Can LLMs Really Attribute Failures in AI Agents?**, NeurIPS 2026.
8. Jiale Liu, **Yifan Zeng**, Shaokun Zhang, Chi Zhang, Malte Højmark-Bertelsen, Marie Normann Gadeberg, Huazheng Wang, Qingyun Wu. **Divide, Optimize, Merge: Fine-Grained LLM Agent Optimization at Scale**, EMNLP 2025 (Findings).
9. Jiahao Qiu*, Yifu Lu*, **Yifan Zeng**, Jiacheng Guo, Jiayi Geng, Chenhao Zhu, Xinzhe Juan, Ling Yang, Huazheng Wang, Kaixuan Huang, Yue Wu, Mengdi Wang. **TreeBoN: Enhancing Inference-Time Alignment with Speculative Tree-Search and Best-of-N Sampling**, EMNLP 2025 (Findings).
10. Haoyu Zheng, **Yifan Zeng**, Linghao Song, Murali Emani, Wenqian Dong. **Enabling Unstructured Sparse Fine-Tuning and Inference for Foundation Models on Wafer-Scale Engine**, SC '25 Workshops.
11. Yaoyao Qian, Yuanli Wang, Jinda Zhang, Yun Zong, Meixu Chen, Hanhan Zhou, Jindan Huang, **Yifan Zeng**, Xinyu Hu, Chan Hee Song, Danqing Zhang. **WebGraphEval: Multi-Turn Trajectory Evaluation for Web Agents using Graph Representation**, NeurIPS 2025 Workshop MTI-LLM.

* denotes equal contribution.